

3D Modelling Fundamentals

Computing and Mathematics

May, 2026



3D Modelling Fundamentals (A13461)

Short Title:	3D Modelling Fundamentals	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Graphic Design and Animation	
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Credits	5	Level: Introductory

Description of Module / Aims

This module serves as an introduction to 3D modelling and comprises an emphasis on the modelling pipeline, from concept to model creation. Students will acquire a knowledge of the craft of 3D modelling and will gain experience in producing a number of 3D models using a number of essential tools within an industry-led 3D-creation toolset.

Programmes

COMP-0595	BSc (Hons) in Creative Computing (WD_KCRC0_B)	2 / 4 / M
COMP-0595	BSc in Multimedia Applications Development (WD_KMULA_D)	2 / 4 / M

Indicative Content

- Introduction to the 3D production pipeline, from concept to mesh creation
- Components of an industry-standard 3D modelling platform
- Polygon, subdivision, and NURB modelling
- Basic camera and viewing techniques
- Introductory shading and rendering

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the fundamentals of the 3D modelling workflow, from concept to a shaded mesh.
2. Identify and effectively navigate the main components and features of an industry standard 3D modelling production platform.
3. Compare various modelling approaches, i.e. polygon, sub-division and NURB-based modelling.
4. Demonstrate the use of appropriate tools for the creation and manipulation of 3D objects and object components.

Learning and Teaching Methods

- Interactive and open-forum lectures.
- Class discussions and presentations.
- Problem-based learning activities.
- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4
<i>Assignment</i>	30%	1
<i>Project</i>	70%	2,3,4

Assessment Criteria

- <40%:** Inability to identify and navigate the main components of a 3D production software platform. Inability to critically evaluate techniques used in the creation and exporting of 3D computer graphics models.
- 40%-49%:** Ability to identify and navigate the main components of a 3D production software platform. Ability to describe key concepts of the 3D modelling pipeline.
- 50%-59%:** Ability to discuss and critically evaluate key concepts of the 3D modelling pipeline and techniques.
- 60%-69%:** Be able to solve problems within the framework of the 3D modelling pipeline by experimenting with appropriate skills and tools of the 3D production software.
- 70%-100%:** All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of problems through the use and modification of appropriate skills and tools of the 3D production software.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Supplementary Material

- Palamar, T. _Mastering Autodesk Maya 2016_. New York: Sybex, 2015.

Requested Resources

- COMPUTER LAB: BYOD Lab